PySR on Fully Observable NeuronBench

A controlled symbolic-regression comparison of vanilla PySR and evolved run 538190

Six deterministic worlds × two methods × three seeds; 10^6 maximum evaluations per fit

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Bottom line. Vanilla PySR produced the only numerically recovered vector field (Na-fatigue, seed 1; held-out NRMSE 3.44×10^{-7}), but neither method achieved literal symbolic identity. Across all 18 paired fits the result was a **9–9 tie**. Evolved 538190 was modestly better on affine-calibrated shape error, as expected from its affine-invariant loss, but it did not improve full physical-dynamics recovery.

1 Question and reduction

NeuronBench[1] normally combines active experiment design, partially observed gating state, and mechanistic model discovery. This study deliberately removes the first two difficulties. For every deterministic world, the regressor observes injected current I_{ext}, membrane voltage V, and every composite channel-open fraction ϕ_c from the benchmark's Eq. (32). The target is the exact voltage vector field

$$\dot{V} = I_{\text{ext}} - \sum_c g_c\,\phi_c\,(V - E_c). \tag{1}$$

The result is six ordinary noiseless scalar SR problems. The composite features remain mechanistic: for example, $\phi_{\text{Na}} = m_{\text{Na}}^3 h_{\text{Na}}$ and $\phi_{\text{K}} = n_{\text{K}}^4$; in Na-fatigue, $\phi_{\text{Na}} = m_{\text{Na}}^3 h_{\text{Na}} s_{\text{Na}}$ includes the slow inactivation state.

Table 1: World-specific term added to the common Na/K/leak current balance.

World	Additional observable / state change	Additional term in $\dot{V}$
Z-rebound	$\phi_Z = m_Z^2 h_Z$	$-4\phi_Z(V-120)$
H-sag	$\phi_h = m_h$	$-5\phi_h(V+30)$
Na-fatigue	ϕ_{Na} includes s_{Na}	none beyond modified ϕ_{Na}
Ca-rebound	$\phi_T = m_T^2 h_T$	$-3.2\phi_T(V-120)$
D-type	$\phi_D = m_D h_D$	$-9\phi_D(V+77)$
Textbook-M	$\phi_M = m_M$	$-2.5\phi_M(V+77)$

2 Experimental protocol

- **Data.** 1,024 training and 16,384 held-out scrambled-Sobol states per world, covering $I_{\text{ext}} \in [-40, 20]$, $V \in [-95, 60]$, and each $\phi_c \in [0, 1]$. Targets are analytic evaluations of Eq. (1)—there is no noise or numerical differentiation.
- **Shared search settings.** Binary operators $\{+, -, \times\}$, no unary operators, 15 populations of 33 members, maximum size 35, Float64 arithmetic, deterministic serial execution, and 10^6 maximum evaluations. Target RMS scaling is inverted before every reported error.
- **Only treatment difference.** Evolved 538190 replaces PySR's mutation, selection, survival, and loss with the saved bundle from `runs/538190`. All data, seeds, and search hyperparameters are otherwise identical.
- **Assessment.** For each run, the reported error is the lowest held-out NRMSE attained anywhere on the discovered Pareto frontier (an oracle assessment of the frontier, not test-set model selection). Numerical recovery means NRMSE $\le 10^{-6}$; near-exact means $\le 10^{-3}$. Exact symbolic equality is checked separately with SymPy.

Table 2: Overall comparison across 18 fits per method. Lower NRMSE is better.

Method	Recovered	Exact symbolic	Median raw NRMSE	Median affine-calibrated NRMSE
Vanilla PySR	1/18	0/18	4.09×10^{-3}	4.02×10^{-3}
Evolved 538190	0/18	0/18	4.63×10^{-3}	3.51×10^{-3}

3 Aggregate results

The raw comparison does not support an overall improvement from 538190. Vanilla has the lower median raw error (4.09×10^{-3} versus 4.63×10^{-3}) and the only strict recovery. The paired win count is exactly 9 evolved to 9 vanilla. The secondary calibrated-shape metric favors 538190 (3.51×10^{-3} versus 4.02×10^{-3}), which is consistent with its custom loss: that loss scores an affine correction $af(x)+b$ internally, while the physical equation must itself recover scale and offset.

Table 3: Best and median raw frontier error over three seeds. "Winner" compares medians.

	GT	Vanilla PySR		Evolved 538190		
World	complexity	best	median	best	median	winner
Z-rebound	31	3.38×10^{-3}	3.46×10^{-3}	3.12×10^{-3}	3.66×10^{-3}	Vanilla
H-sag	31	3.47×10^{-3}	3.55×10^{-3}	1.68×10^{-3}	3.74×10^{-3}	Vanilla
Na-fatigue	23	3.44×10^{-7}	3.05×10^{-5}	1.00×10^{-4}	1.73×10^{-3}	Vanilla
Ca-rebound	31	4.28×10^{-3}	9.79×10^{-3}	3.92×10^{-3}	8.58×10^{-3}	Evolved
D-type	31	4.00×10^{-3}	4.17×10^{-3}	7.86×10^{-4}	5.34×10^{-3}	Vanilla
Textbook-M	31	3.85×10^{-3}	5.35×10^{-3}	6.55×10^{-3}	6.76×10^{-3}	Vanilla

[1] https://github.com/murphyk/neuronbench, pinned at c354622458c460b419cab821d482c879f0578377. The benchmark is introduced in https://arxiv.org/abs/2608.09696.

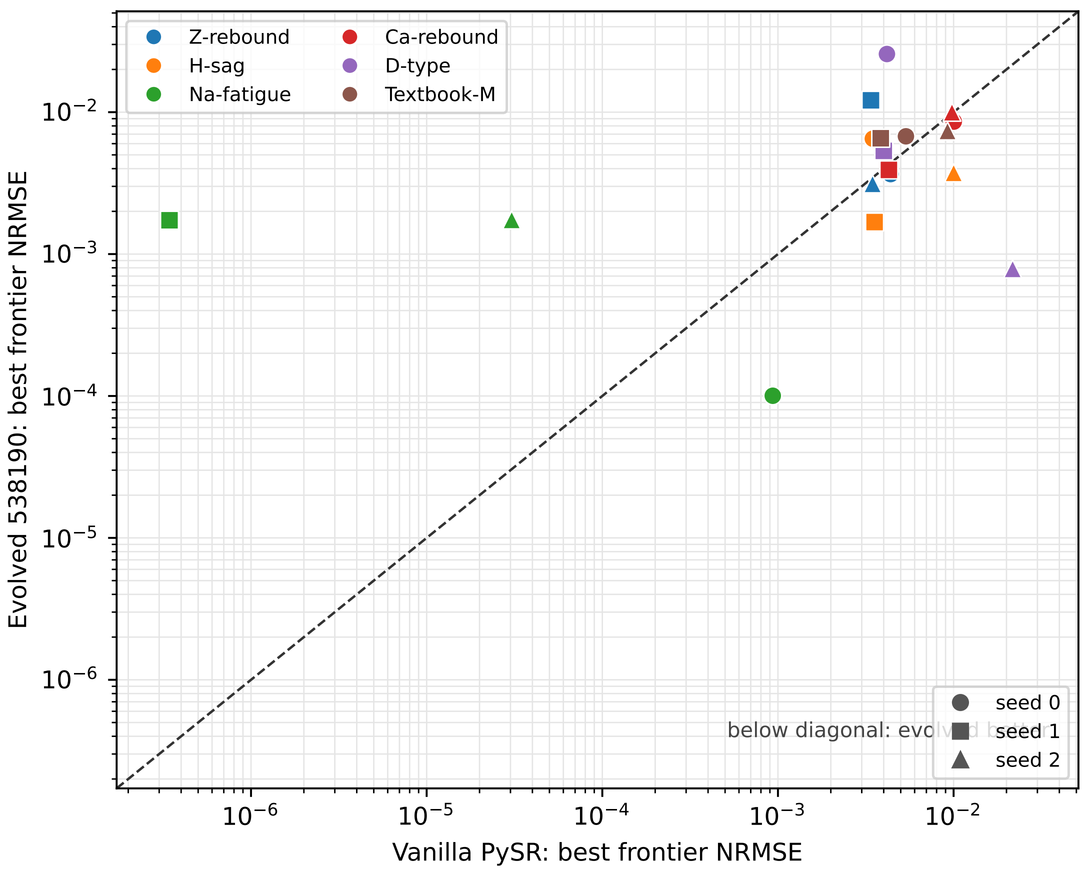

Figure 1: Per-seed comparison of the best equation anywhere on each discovered frontier. Markers encode seed and colors encode world. The 9–9 split around the diagonal shows that neither method is consistently better.

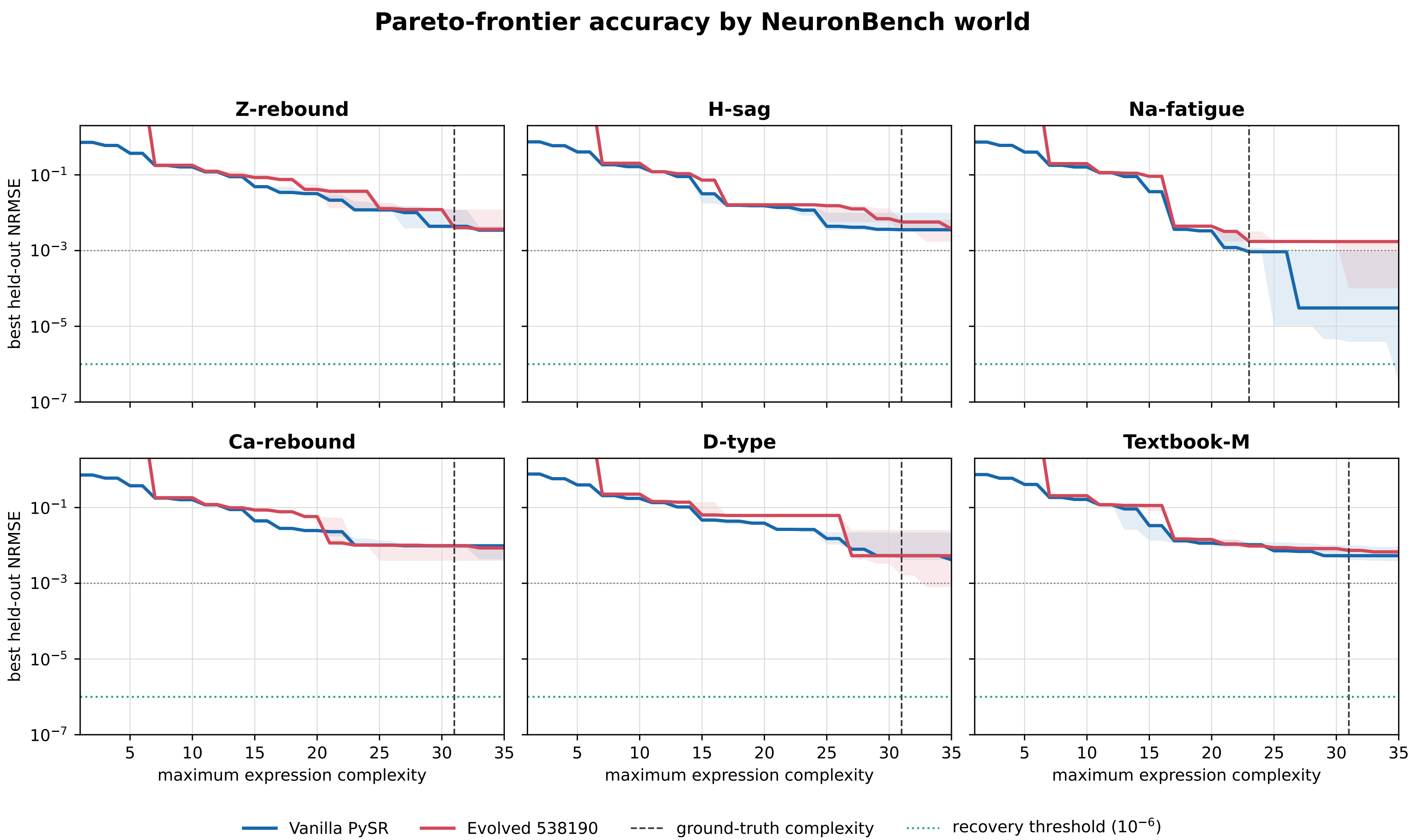

Figure 2: Held-out accuracy along the discovered Pareto frontiers. Solid curves are the median best-so-far envelope over three seeds; shaded bands span the seedwise minimum to maximum. The vertical dashed line is the reference ground-truth complexity. The horizontal teal line is the strict numerical-recovery threshold. Na-fatigue is the only world to cross it.

4 What the one recovered run found

Vanilla PySR's Na-fatigue seed-1 frontier contains a complexity-35 expression with held-out NRMSE 3.436×10^{-7}. It is not literally identical under symbolic simplification, but expanding its redundant tree exposes an extremely accurate coefficient match:

$$\dot{V}_{\text{true}} = I_{\text{ext}} - 36\phi_K V - 2772\phi_K - 120\phi_{\text{Na}} V + 6000\phi_{\text{Na}} - 0.3V - 16.32, \tag{2}$$

$$\begin{aligned}\widehat{\dot{V}} = &\ 1.000045 I_{\text{ext}} - 35.999901\phi_K V - 2771.9985\phi_K - 119.99995\phi_{\text{Na}} V \\ &+ 6000.0037\phi_{\text{Na}} - 0.300082V - 16.322334.\end{aligned} \tag{3}$$

Thus the numerical recovery is scientifically meaningful, although the search used 12 more nodes than the compact reference expression (35 versus 23) and did not reproduce exact floating constants.

Table 4: Expanded coefficients for the recovered Na-fatigue run.

Term	truth	discovered	relative error
I_{ext}	1	1.000045	4.5×10^{-5}
$\phi_K V$	-36	-35.999901	2.8×10^{-6}
ϕ_K	-2772	-2771.9985	5.4×10^{-7}
$\phi_{\text{Na}} V$	-120	-119.99995	4.2×10^{-7}
ϕ_{Na}	6000	6000.0037	6.2×10^{-7}
V	-0.3	-0.300082	2.7×10^{-4}
constant	-16.32	-16.322334	1.4×10^{-4}

5 Best evolved near-miss

The strongest evolved result is D-type seed 2 (NRMSE 7.862×10^{-4}). After expansion,

$$\begin{aligned}\widehat{\dot{V}}_{538190} = &\ 0.997137 I_{\text{ext}} - 9.18239\phi_D V - 695.786\phi_D - 36.1745\phi_K V - 2775.07\phi_K \\ &- 120.193\phi_{\text{Na}} V + 5997.49\phi_{\text{Na}} - 11.5219.\end{aligned} \tag{4}$$

It identifies all three gated-current structures and gets their coefficients close, but it omits the leak slope $-0.3V$ and shifts the constant (truth -16.32). This is exactly the distinction between shape recovery and a correct physical vector field that the raw-versus-calibrated evaluation is meant to expose.

6 Conclusions and limitations

1. **Vanilla PySR can solve at least one simplified NeuronBench world.** It numerically recovers Na-fatigue in one of three seeds, with coefficient errors mostly below 10^{-4} relative.

2. **Run 538190 does not improve strict recovery at this budget.** It has zero strict recoveries, a slightly worse raw median, and an exactly tied paired win count.

3. **538190 does improve affine-calibrated shape modestly.** This aligns with its evolved loss rather than demonstrating better recovery of physically scaled dynamics.

4. **This is intentionally not the original benchmark task.** Composite open fractions are supplied, all states are sampled directly, and the study learns only $\dot{V}$. It does not test active design, recovery of gate ODEs, latent-state inference, or trajectory forecasting.

5. **Three seeds are enough for a demo, not a fine-grained ranking.** The broad seed bands and 9–9 paired split argue against claiming a small algorithmic advantage either way.

7 Appendix: all 36 best-frontier errors

Table 5: Held-out raw NRMSE of the closest equation anywhere on each run's frontier. Green marks strict recovery; blue marks near-exact ($\leq 10^{-3}$).

	Vanilla PySR			Evolved 538190		
World	seed 0	seed 1	seed 2	seed 0	seed 1	seed 2
Z-rebound	4.38×10^{-3}	3.38×10^{-3}	3.46×10^{-3}	3.66×10^{-3}	1.21×10^{-2}	3.12×10^{-3}
H-sag	3.47×10^{-3}	3.55×10^{-3}	1.00×10^{-2}	6.50×10^{-3}	1.68×10^{-3}	3.74×10^{-3}
Na-fatigue	9.34×10^{-4}	3.44×10^{-7}	3.05×10^{-5}	1.00×10^{-4}	1.73×10^{-3}	1.74×10^{-3}
Ca-rebound	1.00×10^{-2}	4.28×10^{-3}	9.79×10^{-3}	8.58×10^{-3}	3.92×10^{-3}	9.99×10^{-3}
D-type	4.17×10^{-3}	4.00×10^{-3}	2.17×10^{-2}	2.57×10^{-2}	5.34×10^{-3}	7.86×10^{-4}
Textbook-M	5.35×10^{-3}	3.85×10^{-3}	9.23×10^{-3}	6.76×10^{-3}	6.55×10^{-3}	7.38×10^{-3}

Reproducibility

NeuronBench was installed into the `meta_sr` conda environment at the pinned commit above. The complete frontiers and metrics are in `reports/neuronbench_pysr_fully_observable.results.json`. The experiment runner is `scripts/neuronbench_fully_observable.py` and this report is regenerated by `scripts/build_neuronbench_latex_report.py`. SLURM array job 142502 completed all 36 tasks without a nonzero exit or traceback.